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Career Longevity and Performance Following Shoulder Instability in National Football League Athletes

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Career Longevity Following Shoulder Instability Surgery in NFL Players

Abstract

Purpose: To investigate the career longevity, game utilization and performance of National Football League (NFL) athletes following glenohumeral instability events treated operatively versus nonoperatively.

Methods: Public resources identified NFL players who sustained a shoulder instability event from 2000-2019. Players with prior shoulder instability, without NFL experience before injury, or who did not return to play (RTP) after injury were excluded. Demographic information, utilization (games and seasons), and season approximate value (SAV) statistics were recorded one year prior to injury and three years following return to play (RTP). Statistical analysis compared utilization and SAV following RTP for athletes managed operatively versus nonoperatively.

Results: Ninety-seven NFL players who sustained their first instability event while playing in the NFL were identified, 91 of whom RTP (93.8%). Quarterbacks were significantly more likely to undergo immediate surgical management compared to other positions (P=.023). Final analysis included 58 players managed operatively and 33 players managed nonoperatively by the end of the index season. Players treated operatively played in significantly more seasons following RTP during their remaining career (4.1±2.7 vs. 2.8±2.5 seasons; P=.015). There were no differences in games played or started, offensive or defensive snap count percentage, or performance (SAV) before and after injury when compared between cohorts (P>.05). Following surgical stabilization, time to RTP (36.62±10.32 vs. 5.43±12.33 weeks, P<.05) and time interval before recurrent instability (105.7±100.1 vs. 24.7±40.6 weeks, P<.001) were significantly longer than with non-operative treatment. Additionally, the operative cohort experienced less recurrent instability (27% vs. 50%; P=.035).

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Conclusions: Athletes who RTP in the NFL following a shoulder instability injury do so with similar workload and performance irrespective of surgical or non-surgical management. While nonoperative treatment is associated with faster return to play, operative management is associated with fewer recurrent instability events, greater time between recurrent instability events, and greater career longevity.

Level of Evidence: Level III; Retrospective Case-Control Study.

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Introduction

In the United States, glenohumeral joint instability has a reported incidence of 23.9 cases per 100,000 person-years.[1] High school and collegiate athletes demonstrate increased incidence of these injuries,[2,3] so it is no surprise that the prevalence of shoulder instability is reported to be as high as 10-14% among National Football League (NFL) athletes.[4,5] In an investigation of shoulder injuries in elite football players invited to the NFL Combine, it has been shown that 24% of all shoulder injuries in this population were due to instability events, with defensive players making up the greatest proportion, requiring surgical management 76% of the time.[4] In a retrospective cohort study of 328 NFL players who started their career on a single NFL team from 1980 to 2008, 9.5% of players had a previous history of shoulder instability and 13.1% of players experienced their first shoulder instability injury at a mean of 18.4 ± 22.2 months after joining the team.[6] With only 2.3% of NFL games reported as being injury-free, the physical demands of the sport and risk of these injuries remain an important considerations for NFL athletes.[7,8]

A number of studies have evaluated the impact of shoulder instability in elite football players. Okoroha et al demonstrated high levels of return to play (RTP) following shoulder instability injuries among NFL athletes, with no difference in career longevity between operative and nonoperative groups,[9] while Brophy and colleagues found that offensive and defensive linemen demonstrated significantly shorter careers.[10] Prior studies have also shown that players managed nonoperatively experience earlier and more frequent reinjuries compared to those managed surgically.[6,9] Surgical stabilization in the elite athletic population has been shown to reduce the rate of recurrent instability and improve player satisfaction.[6,11-15] Nevertheless, the

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effect of these injuries on player performance, as measured by SAV, following RTP is not known.

While prior investigations have examined the effects of operative and nonoperative management of glenohumeral instability on RTP and incidence of recurrent instability events in NFL players, there is currently a paucity of literature examining game utilization and performance under these conditions. Additionally, subsequent management following recurrent instability events is not well described following initial treatment in NFL athletes. The purpose of this study is to investigate the career longevity, game utilization and performance of NFL athletes following glenohumeral instability events treated either operatively or nonoperatively. The authors hypothesize that upon RTP, NFL athletes who underwent surgical stabilization will have increased career longevity, game utilization and performance when compared to athletes managed nonoperatively.

Methods

All incidences of shoulder instability among NFL players during a twenty-year period from September 2000 to February 2019 were collected. These players were found using public internet searches similar to previous studies.[16-24] The internet searches utilized sources such as team websites, team injury reports, and sports news websites to determine a comprehensive list of players who sustained a shoulder instability event during their NFL career. The reports were cross-referenced with multiple sources to ensure their accuracy and correlated to when athletes missed games. The terms “shoulder dislocation”, “shoulder subluxation”, “shoulder instability”, and “glenohumeral joint instability,” and “labrum injury” were accepted terms in the references

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as documentation of a shoulder instability event, which allowed for inclusion in this study. These searches were conducted by a research team in the Sports Medicine division of the Orthopedic Surgery Department, consisting of MD candidates and orthopedic surgery residents, and further verified by faculty. All identified players underwent further internet searches to determine whether they are included or excluded based on the following criteria. Inclusion criteria consisted of all NFL players who had a shoulder dislocation or subluxation during their NFL careers reported over the prior two decades. Players were excluded from the study if they did not play in the NFL one season prior to the index (injury) season, if they did not return to play following their injury, if they had a history of prior ipsilateral upper extremity surgery, or if their return to play data was not complete for an entire season for analysis.

Publicly available sources were used to record demographic information including age, body mass index (BMI) and position, at the time of injury – the index season. The index season was defined as the NFL season (and subsequent offseason) during which a player sustained their first instability event. Therefore, the season prior to the index season was defined as season -1 and the first season following injury in which a player returned to play was defined as season +1. Based on the index season designation, game utilization parameters and performance metrics (SAV) from three seasons prior to (years -1, -2, -3) and following (years +1, +2, and +3) the index year were analyzed. If a player sustained a recurrent injury after season +1, data collection on that player was ceased after their recurrent injury as to limit bias in their performance metrics not attributable to the index injury. Game utilization parameters documented included seasons played, games played (GP), and games started (GS) per season, as well as offensive snap count percentage (OSCP) and defensive snap count percentage (DSCP). Performance measurements were documented as the season approximate value (SAV) calculation, which takes into account

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several different correlates to quantify the relative contribution a player made to his team's success. The calculation of SAV is dependent on the position an athlete plays, so there is no single formula for all players. However, SAV has been reported in the literature as a normalized metric that is comparable across positions and standardized to correlate to each player's contribution to their team's success.[7] Additionally, injury specific data points were recorded for individual instability events. These include quarter of season of injury, RTP time (in weeks), subluxation or dislocation, treatment, the presence recurrent instability following the injury, and the interval time from the previous injury until the recurrent instability.

Players were categorized into an operative or non-operative cohort based on final treatment status at the end of the index season (Figure 1). If a player was treated operatively following their initial instability event, as determined by news reports and cross-referenced with an absence of in-game data during the recovery time period, they were placed in the operative cohort. Additionally, players who were treated non-operatively following their initial instability event and did RTP during the index season, but subsequently either sustained a recurrent injury requiring surgery or elected to undergo surgical stabilization during the offseason following the index year, were crossed-over into the operative cohort. Those who had delayed surgery following the index season are designated by "crossed groups" in Figure 1. All other players who were initially treated non-operatively and did not have surgery at the conclusion of the index seasons remained in the non-operative cohort. This designation of operative and non-operative cohorts allowed comparisons of cohorts in subsequent seasons to account for whether a player ultimately underwent surgery. Statistical analysis compared career statistics, such as game utilization and performance, before and after the index season between the operative and non-operative cohorts.

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Statistical Analysis

All continuous data are reported as Mean ± Standard Deviation, while categorical data are reported as counts and column percentages (N (%)). For continuous variables, univariate two-group comparisons are performed using independent 2-sample t-tests if the variable is normally distributed and using Wilcoxon rank sum tests if the variable is non-normally distributed. For categorical variables, univariate two-group comparisons are performed using chi-square tests when expected cell counts are >5 and using Fisher's exact tests when expected cell counts are <5. Repeated measures were performed to see if the performance variables changed differently over time between groups, by comparing seasons +1, +2 and +3 relative to season -1. For repeated measures analyses, data are reported as Adjusted means (standard errors). Additionally, relative percentages of each post-RTP seasons were calculated using season -1 as the baseline, and compared between cohorts. Pearson partial correlations and one-way ANOVA are used to assess if demographics factors are associated with workload and performance variables. Statistical significance is set at $p<0.05$. All analyses are performed using SAS 9.4 (SAS Institute Inc., Cary, NC, USA).

Results

Demographics

There were 97 players identified who experienced a first-time shoulder instability event. After the index instability event, 37 (38.1%) players were initially managed operatively, while 60 (61.9%) players elected for a trial of conservative management. When comparing initial treatment choice by player position, a significantly greater proportion of quarterbacks (22% vs.

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7%) were treated operatively, while a significantly lower proportion of position players (cornerbacks - 11% vs 22%, safeties - 11% vs 20%, and running backs - 5% vs 15%) were treated conservatively, respectively (P = .023**).**

Figure 1 illustrates the composition of the final cohorts following the index season. There were 4 players from the operative cohort who did not RTP and were excluded from analysis. Additionally, of the 60 players who initially chose non-operative treatment, 2 did not RTP and 25 elected to undergo surgery in the offseason, and were therefore crossed-over from the non-operative cohort to the operative cohort . Therefore, final analysis for game utilization and performance following RTP included 58 players treated operatively and 33 players treated non-operatively by the end of the index season. Table 1 outlines the comparison of demographic characteristics between the final operative and non-operative cohorts. The was no statistical difference in age between the operative and non-operative cohorts (25.7 ± 2.6 and 27 ± 3.9, P=.09). No other demographic differences existed between the two cohorts, including position played or time of injury (Table 1).

All players who were treated operatively by the end of their index season were compared to the non-operative cohort to evaluate player workload (GP, GS, OSCP, and DSCP) and performance (SAV) during the first three seasons following the index season (+1, +2, and +3) and remaining career (Table 2). There were no workload or performance differences between operative and nonoperative cohorts in the season prior to injury (season -1) or the season following RTP (season +1). Additionally, no workload or performance differences existed between the two cohorts during the first 3 seasons following RTP relative to the three seasons prior to injury (P>.05). Athletes treated operatively played significantly more seasons following RTP than those treated nonoperatively (4.1 ± 2.7 vs. 2.8 ± 2.5 seasons, P = .02), however they

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played a similar number of total games (30.21 ± 13.47 vs. 28.57 ± 14.92, P>.05) and averaged a similar amount of games per season (11.05 ± 4.13 vs. 12.70 ± 3.58, P>.05) during their remaining career. Athletes treated operatively and nonoperatively performed at similar proficiency as measured by SAV after returning to play at all time points (P>.05). Furthermore, athletes in both cohorts averaged a similar number of Probowl selections before and after injury (P=.15 and P=.92, respectively), indicating similar performance and value to their teams (Table 3).

There were no differences in change of production of GP, GS, OSCP, DSCP, or SAV during any seasons relative to preinjury baseline when compared between operative and non-operative cohorts (P>.05). Additionally, relative percentages were calculated using the season prior to injury as the 100% benchmark. Workload and performance relative percentages during each of the first three seasons following RTP were similar between both cohorts (P>.05), as illustrated in Table 4.

There was a total of 40 recurrent instability events sustained by 38 different players identified. Of the 38 players with recurrent instability events, 36 had only one recurrent episode, and 2 had two recurrent episodes. Recurrent instability events were more prevalent in athletes who opted for initial conservative management (50% vs. 27%, P = .035) immediately following the initial injury. Additionally, operatively managed players had a significantly longer time interval prior to recurrent instability than those treated nonoperatively (105.7 ± 100.1 versus 24.7 ± 40.6 weeks, P < .001). Following an instability event, athletes treated non-surgically averaged a RTP time of 5.43 (95% CI 2.72-8.13) weeks, while those treated surgically averaged a RTP time of 36.62 (95% CI 33.44-39.79) weeks.

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Discussion

The present investigation determined that there is no difference in performance between operative and nonoperative cohorts following a shoulder instability injury in NFL athletes. Additionally, game utilization following RTP was also similar between operative and nonoperative cohorts. Players treated nonoperatively were able to RTP quicker; however, those treated operatively had reduced recurrence rates, longer intervals between recurrent injuries, and longer careers. Glenohumeral instability events are common injuries among football players,[4,5] with evidence of high RTP rates and reduced recurrence after surgical stabilization.[9] The present study evaluated the effects of operative versus nonoperative management of these injuries on player game utilization, performance, and career longevity following RTP, finding similar outcomes between both treatment cohorts.

Several studies have examined the performance of athletes following RTP from shoulder instability. In a retrospective cohort study, Lu et al. evaluated the athletic performance of 50 National Basketball Association (NBA) athletes following RTP from shoulder subluxation or dislocation injuries.[25] They found that following RTP, NBA athletes saw a significant reduction in their Player Efficiency Rating (PER) from 14.8 preinjury to 13.6 postinjury regardless of treatment type. Li et al. also investigated PER in 60 NBA players following RTP from glenohumeral instability events and found that players undergoing nonoperative management had significantly decreased PER compared to players managed surgically.[26] The present study evaluated performance following RTP from shoulder instability in NFL players. There was no difference in performance (SAV) between players managed operatively and nonoperatively following RTP, which is in contradiction to observations made in NBA players. This is most likely due to the unique physical demands of basketball players being heavily reliant on overhead

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activities, such as rebounding and shooting, which are metrics weighed into the PER calculation.[27] Conversely, the largest portion of our cohort was comprised of cornerbacks and safeties, who sustain injuries primary from tackling and are less reliant on overhead activities.[28] These findings suggest that football players do not experience a decline in performance following a shoulder instability event.

Player utilization following RTP has been previously evaluated in football players at the collegiate level. In a case series by Robins and colleagues evaluating National Collegiate Athletic Association Division I football players, the number of games played following RTP was found to increase by 20% when compared to preinjury in operatively treated players.[29] They illustrated that college players undergoing surgical stabilization may RTP at the same or higher levels of utilization. In a cohort study by Li et al investigating shoulder instability among NBA athletes, it was demonstrated that players who underwent shoulder surgical stabilization maintained the number of games played, while players treated nonoperatively saw significant reductions in games played.[26] Conversely, the present findings suggest that there is no significant difference in game utilization following RTP between surgically and non-surgically managed players following shoulder instability. Additionally, a large portion of players were initially treated nonoperatively and were able to RTP during the same season but later crossed over to the operative group by the offseason, without detriment to their utilization and performance the following seasons.

The value of operative management of shoulder instability has been demonstrated in NFL players with consideration of RTP rates, reinjury rates, and career longevity. LeClere et al. performed a retrospective cohort study of 31 NFL players from 1980 to 2008 who sustained shoulder instability events before their professional career and found that players treated

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surgically had a significantly lower rate of recurrence compared to the nonoperative group (13.1% vs 41.7%, P=0.02).[6] Players in the surgical group were also found to have longer intervals between recurrent injuries (18.4 vs. 4.4 months, P=0.04). Okoroha et al. compared the rate of recurrent injury in NFL players managed surgically and nonsurgically and reproduced these findings, with operatively managed players displaying significantly lower rates of recurrence (26% vs. 55%, P=0.021) and longer intervals between recurrent injuries (14.7 vs. 2.5 weeks, P=0.05).[9] The results of the present cohort, investigating players who sustained their first shoulder instability event during their NFL career, echoed the prior studies with reduced recurrence rates in the operative cohort (27% vs. 50%) and longer intervals between recurrent instability events (105.7 ± 100.1 versus 24.7 ± 40.6 weeks). Additionally, players who RTP successfully following operative management had significantly longer careers. Interestingly however, they did not play in a greater number of games as would be expected. This may be the result of non-quantifiable factors, such as coaching decisions, team dynamics, and player factors such as minor injuries or trades, which the present study is unable to account for. Nevertheless, these results suggest the potential career prolongation in NFL athletes who seek surgical stabilization of shoulder instability.

When managing the care for professional football players, it is paramount to recognize that successful RTP does not necessarily equate to satisfactory outcomes. Glenohumeral dislocation and subluxation injuries can be extremely detrimental for NFL athletes as the ability to perform football-specific tasks, such as overhead throwing, blocking/pushing opposing players, or bracing during a fall to the ground,[7,30] can increase the risk of subsequent injury.[31] Additionally, these injuries result in a significant amount of missed game time due to the need of extended rehabilitation.[32] Given the low retention rates and short career lengths in the NFL, with

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average career length reported to have decreased dramatically to as low as approximately 2.66 seasons,[33] NFL players remain dependent on a high level of in-game performance and consistency. Therefore, career longevity is an important factor for an athlete when considering future contract negotiations and estimated earning potential.[34]

There are many other factors to be considered when determining a treatment algorithm. These must take into account injury factors, such as bone loss, engaging lesions, recurrent subluxation and athlete symptoms. Additionally, career factors may drive an athlete's preference on wanting to RTP the same season versus undergoing surgery, such as the team pursuing the playoffs or the player negotiating a new contract. The present investigation elucidates that operatively treated players have reduced recurrent instability injuries and non-operatively treated players RTP quicker; however, neither treatment shows superiority in terms of player performance and game utilization. As such, treatment following a shoulder instability event should be individualized to the athlete's injury characteristics, symptoms, and personal or career preferences.

Limitations

Due to the nature of the retrospective design of this study, there are several limitations. First, there was no priori power analysis conducted. Using publicly available data is a limitation as well, as it is possible that some players were not identified. Additionally, all data was pooled from independent third-party sources, which lack availability of official medical records. This leads to possible inaccuracies, such as recording official RTP clearance as this study could only identify when players returned to actual game play. Additionally, this limits the identification of concomitant injuries or sequalae in the athletes, such as recurrent shoulder subluxations, bone

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loss, and on-track and off-track lesions. Subsequently, the use of third-party sources has limited access to information regarding the type of instability experienced by the players, including hand dominance, mechanism of injury, and the ability to differentiate between anterior, posterior, and superior labral tears. The public resources may not effectively distinguish between first-time and recurrent shoulder instability, as press documentation may lack diagnostic precision. Operative information such as surgical experience of the surgeon, the type of surgery performed, approaches used, and implants placed, could not be collected or adjusted for. Additionally, injuries involved two decades of NFL play, treatments, and rehabilitation protocols, allowing for variation in management amongst players. Furthermore, it is not possible to account for changes in coaching and roster composition over the last two decades that can impact player utilization and performance. It should also be noted that players who sustained injuries prior to their professional career may not have had available internet records and as such may not have been identified as appropriately by the exclusion criteria.

Conclusion

Athletes who RTP in the NFL following a shoulder instability injury do so with similar workload and performance irrespective of surgical or non-surgical management. While nonoperative treatment is associated with faster return to play, operative management is associated with fewer recurrent instability events, greater time between recurrent instability events, and greater career longevity.

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Figure Legends

Figure 1. Flow diagram of inclusion of operative and non-operative cohorts.

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Table 1. Demographics of NFL players with a Shoulder Instability Event

		Following Index Season		
Variable	**Response**	**Operative (N = 58)**	**Non-Operative (N = 33)**	**p-value**
Age	Mean ± STD	25.71 ± 2.55	27.03 ± 3.87	0.085
BMI	Mean ± STD	29.38 ± 3.58	29.19 ± 3.27	0.804
Position	QB	8 (14%)	3 (9%)	0.255
	WR	8 (14%)	3 (9%)	
	G	3 (5%)	2 (6%)	
	S	8 (14%)	8 (24%)	
	LB	6 (10%)	3 (9%)	
	OT	2 (3%)	0 (0%)	
	CB	14 (24%)	3 (9%)	
	TE	1 (2%)	3 (9%)	
	RB	4 (7%)	6 (18%)	
	DE	1 (2%)	2 (6%)	
	DT	1 (2%)	0 (0%)	
	LS	2 (3%)	0 (0%)	
Time of Injury	Preseason - Week 4	19 (33%)	19 (58%)	0.122
	Week 5-8	9 (16%)	5 (15%)	
	Week 9-12	10 (17%)	5 (15%)	
	Week 13 Postseason	19 (33%)	4 (12%)	
	Off season	1 (2%)	0 (0%)	
S or D	Dislocation	53 (91%)	28 (85%)	0.487
	Subluxation	5 (9%)	5 (15%)	

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Continuous Variables are presented using Mean ± SD
Categorical variables are presented using frequency (percentage)
Age represented by years; BMI represented in kg/m2; Time of injury represented by weeks
BMI, body mass index; QB, quarterback; WR, wide receiver; G, guard; S, safety; LB, linebacker; OT, offensive tackle; CB, cornerback; TE, tight end; RB, running back; DE, defensive end; DT, defensive tackle; LS, long snapper; S or D, Subluxation or Dislocation
Significant Values are noted in **bold**, $P<0.05$

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Table 2. Workload and Performance Following a Shoulder Instability Injury

	Season -1			Season +1				Three Season Average (+1, +2, +3)		
Workload Variable	Operative	Nonoperative	P-value	Operative	Nonoperative	P-value	Variable	Operative	Nonoperative	P-value
GP	14.31 ± 0.65	14.63 ± 0.9	1.000	12.4 ± 0.64	13.57 ± 0.88	0.996	Pre-Injury	13.96 ± 0.43	14.24 ± 0.56	0.254
							Post-Injury	11.6 ± 0.4	13.05 ± 0.63	
GS	9.08 ± 0.89	11.67 ± 1.24	0.871	6.75 ± 0.89	9.96 ± 1.22	0.602	Pre-Injury	9.02 ± 0.58	11.42 ± 0.77	0.701
							Post-Injury	6.91 ± 0.55	8.77 ± 0.86	
OSCP	0.3938 ± 0.081	0.404 ± 0.102	1.000	0.1313 ± 0.067	0.335 ± 0.086	0.781	Pre-Injury	0.3830 ± 0.068	0.3691 (0.0697)	0.143
							Post-Injury	0.1241 ± 0.040	0.2865 ± 0.068	
DSCP	0.1888 ± 0.084	0.239 ± 0.107	1.000	0.2604 ± 0.070	0.2292 ± 0.094	1.000	Pre-Injury	0.1506 ± 0.060	0.2478 ± 0.071	0.547
							Post-Injury	0.2666 ± 0.042	0.2876 ± 0.075	
Performance Variable	Operative	Nonoperative	P-value	Operative	Nonoperative	P-value	Variable	Operative	Nonoperative	P-value
SAV	5.69 ± 0.61	6.78 ± 0.84	0.996	4.17 ± 0.6	5.62 ± 0.85	0.967	Pre-Injury	5.71 ± 0.4	6.63 ± 0.52	0.934
							Post-Injury	4.13 ± 0.38	4.96 ± 0.6	

Continuous Variables are presented using Mean ± SD

GP, games played; GS, games started; OSCP, offensive snap count percentage; DSCP, defensive snap percentage; SAV, season approximate value

Significant Values are noted in **bold**, P<0.05

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Table 3. Career Workload and Performance Following Injury

	Remaining Career Post Injury		
Workload	Operative	Nonoperative	P-value
Total Seasons	4.06 ± 2.65	2.82 ± 2.51	**0.015**
Total Games	30.21 ± 13.47	26.57 ± 14.92	0.269
GPS	11.05 ± 4.13	12.70 ± 3.58	0.071
Performance	Operative	Nonoperative	P-value
Probowl Preinjury	0.52 ± 1.25	1.45 ± 2.72	0.150
Probowl Postinjury	0.57 ± 1.98	0.36 ± 1.11	0.917

Continuous Variables are presented using Mean ± SD

GPS, games per season; Probowl pre, probowl selections before injury; Probowl post, probowl selection after injury

Significant Values are noted in **bold**, P<0.05

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Table 4. Mean Differences and Relative Percentages of Workload and Performance Each Season Following Injury Relative to the Season Prior to Injury

		Mean Difference From Season -1			Relative Percentage to Season -1		
Variable	**Season**	**Operative (N = 58)**	**Non-Operative (N = 33)**	**p-value**	**Operative (N = 58)**	**Non-Operative (N = 33)**	**p-value**
GP	Season +1	-1.87 ± 5.72	-0.71 ± 6.20	0.433	1.04 ± 1.12	1.24 ± 1.49	0.374
	Season +2	-1.95 ± 5.93	-1.29 ± 7.30	0.729	1.03 ± 1.06	1.36 ± 1.81	0.906
	Season +3	-3.91 ± 6.96	-3.82 ± 7.39	0.972	0.98 ± 1.39	1.12 ± 1.52	0.989
GS	Season +1	-1.53 ± 6.34	-0.08 ± 7.40	0.393	0.76 ± 0.69	1.55 ± 2.70	0.073
	Season +2	-1.32 ± 7.09	-1.93 ± 6.50	0.776	0.89 ± 0.87	1.05 ± 0.82	0.474
	Season +3	-2.25 ± 8.88	-6.91 ± 5.43	0.11	0.83 ± 0.84	0.35 ± 0.27	0.128
OSCP	Season +1	-0.12 ± 0.31	-0.09 ± 0.42	0.837	0.61 ± 0.61	1.00 ± 0.66	0.337
	Season +2	-0.12 ± 0.36	-	-	0.80 ± 0.90	-	-
	Season +3	-0.15 ± 0.32	-	-	0.44 ± 0.53	-	-
DSCP	Season +1	-0.00 ± 0.29	-0.02 ± 0.21	0.632	3.51 ± 5.97	1.86 ± 2.15	0.868
	Season +2	-0.04 ± 0.35	0.02 ± 0.08	0.585	3.95 ± 8.24	1.29 ± 0.46	0.405
	Season +3	0.03 ± 0.40	-	-	6.12 ± 10.92	-	-

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SAV	Season +1	-0.85 ± 3.79	-1.05 ± 5.13	0.859	1.21 ± 1.57	1.44 ± 1.87	0.797
	Season +2	-1.02 ± 3.99	-2.64 ± 3.77	0.31	1.33 ± 2.43	0.87 ± 0.75	0.828
	Season +3	-1.58 ± 5.50	-4.40 ± 5.48	0.166	1.50 ± 3.28	0.45 ± 0.38	0.249

Repeated measures are presented as Adjusted Mean ± Standard Error

GP, games played; GS, games started; OSCP, offensive snap count percentage; DSCP, defensive snap count percentage; SAV, season approximate value

Significant Values are noted in **bold**, P<0.05

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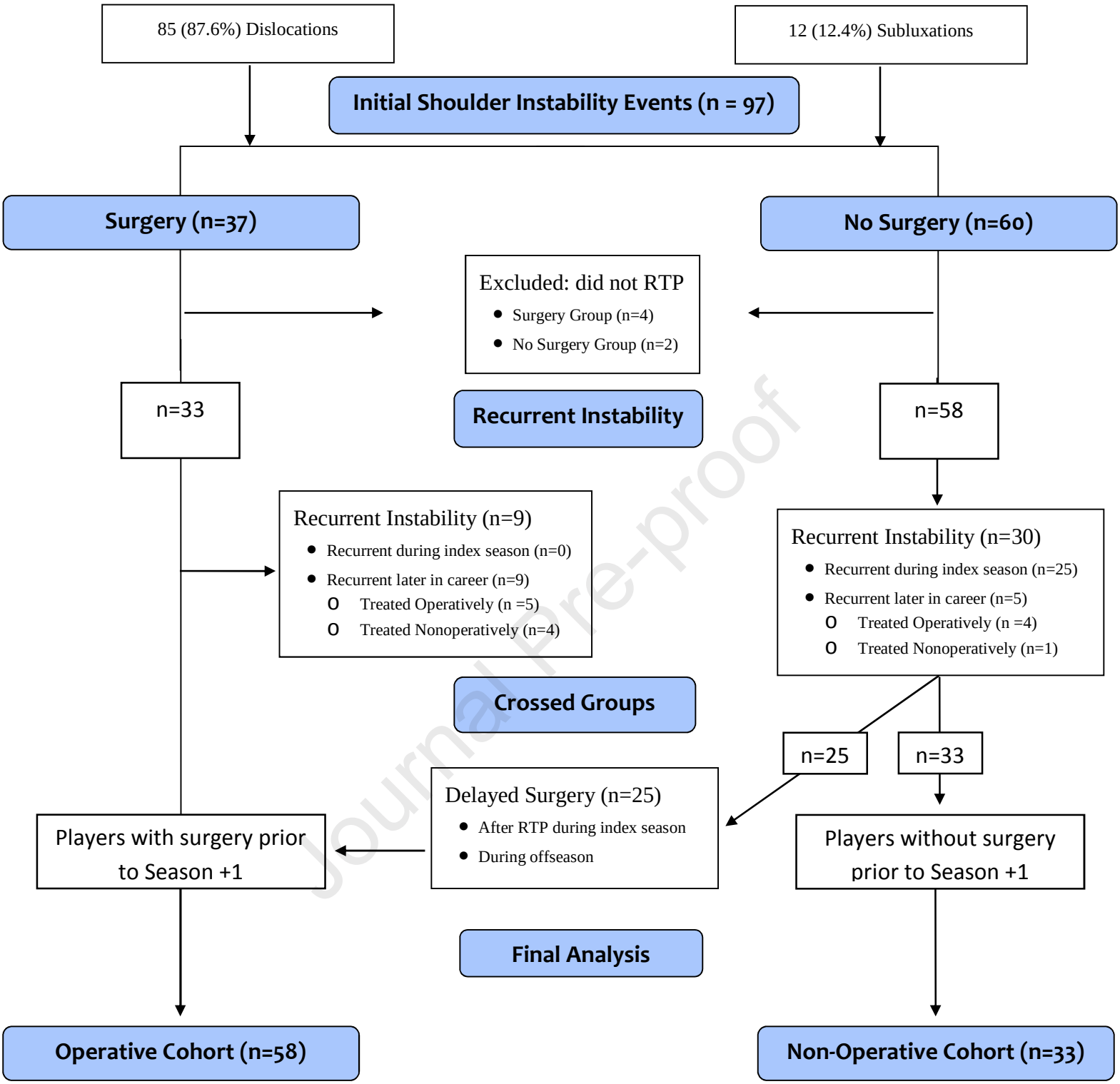

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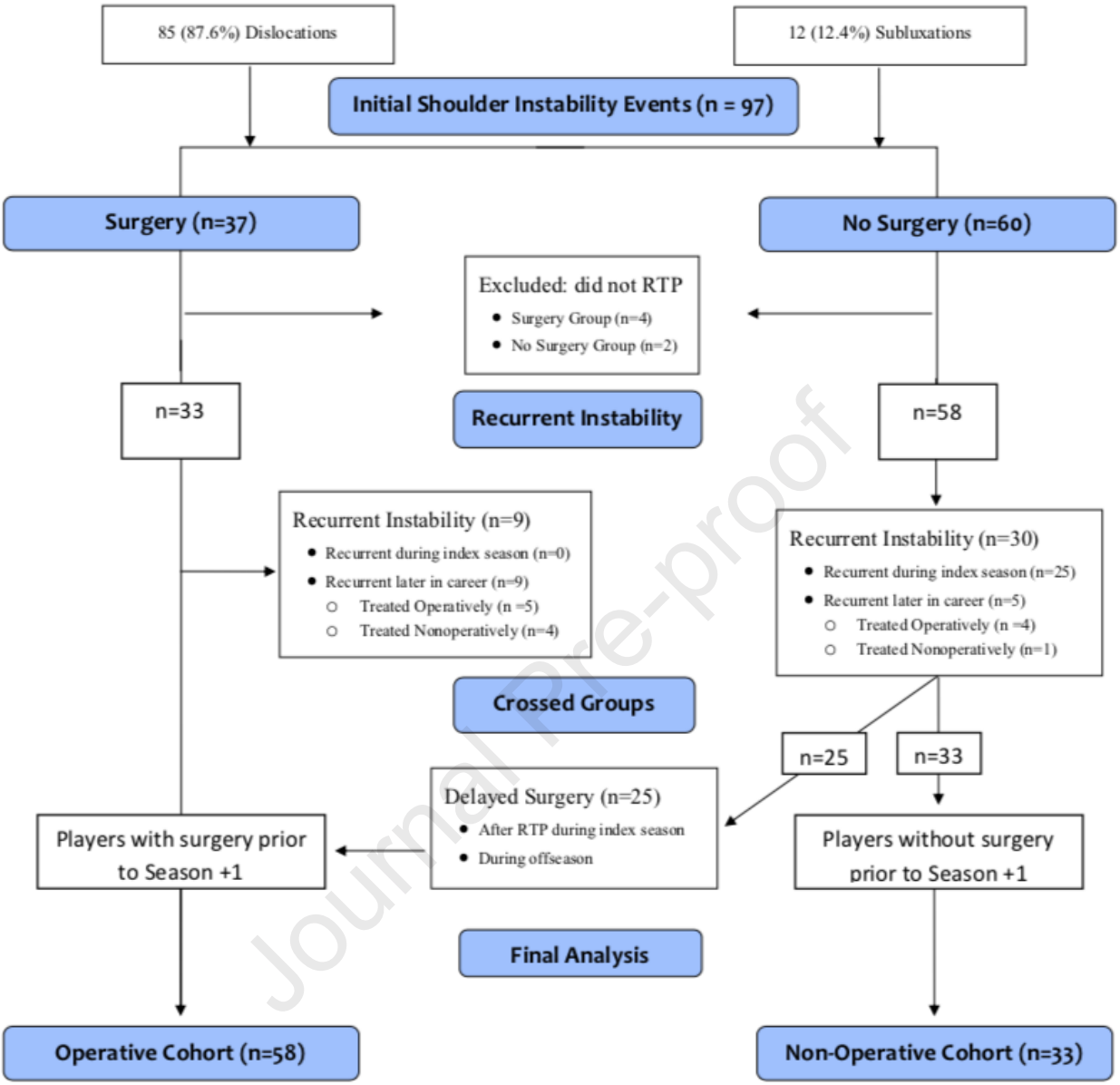

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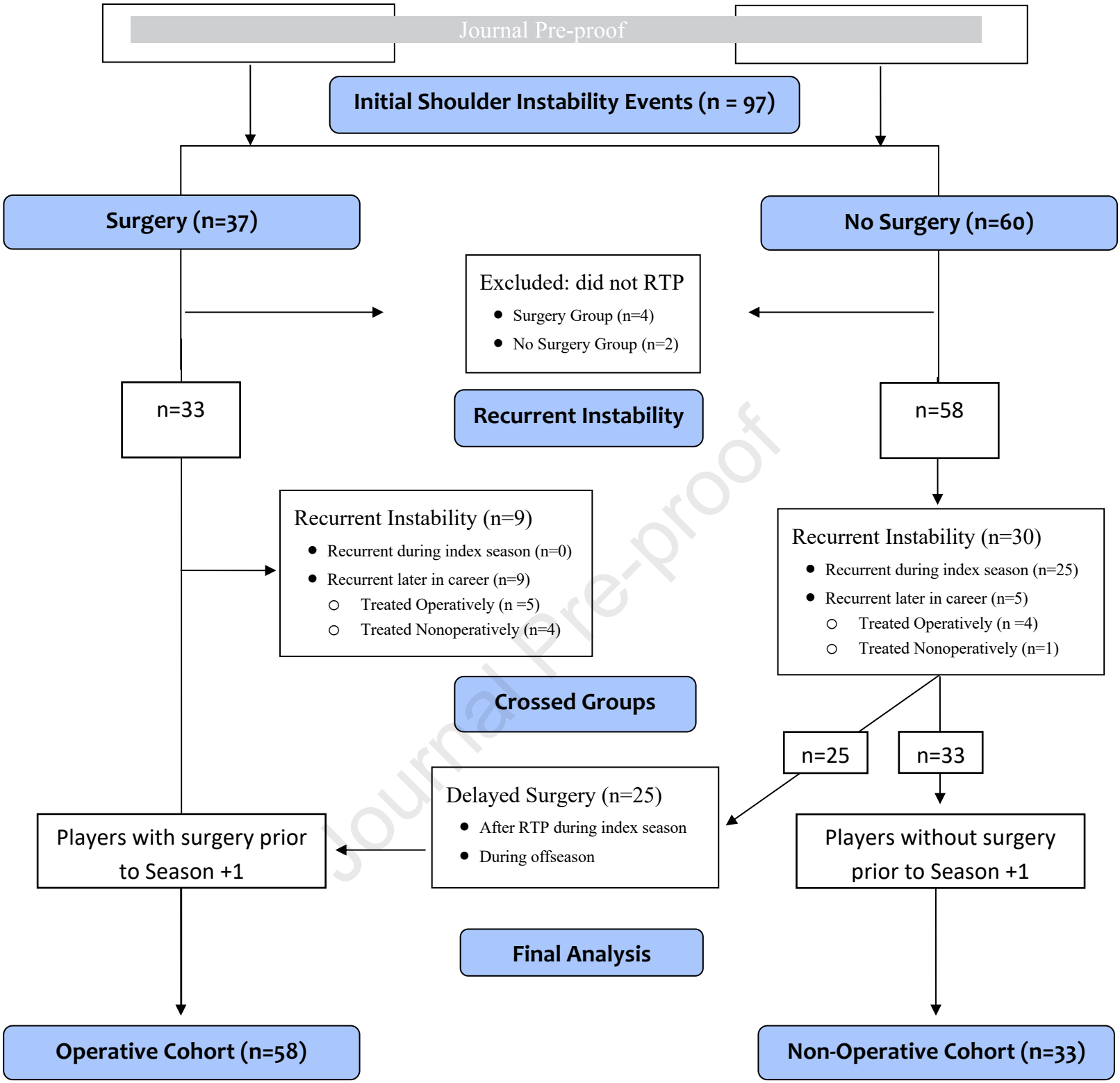

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